



JSPM's
JAYAWANTRAO SAWANT POLYTECHNIC,
Handewadi Road, Hadapsar, Pune-28
Department of Computer Engineering
Academic Year 2025-26



MICRO PROJECT

E-commerce Website Testing

Program: CO

Program code: CO6K

Course: Software Testing

Course code: 316314

Class: TYCO1

Group No: G

Project Guide: Ms.S.P.Sarode



MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION

Certificate

This is to certify that Ms. Sarah Shaikh, Ms. Siddhi Darekar, Ms. Pradnya Kaurav, Ms. Subiya Sayyed. Roll No. 2, 10, 21, 66 of VI Semester of Diploma in Computer Engineering of Institute JAYAWANTRAO SAWANT POLYTECHNIC (Code: 0711) has completed the Micro Project satisfactorily in Subject Software Testing (316314) for the academic year 2025- 2026 as prescribed in the curriculum.

Place: Hadapsar, Pune.

Enrollment No: 2207110109
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23211830431

Date:.....

Exam Seat No : 208454
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Subject Teacher

Head of the Department

Principal

PROJECT GROUP
DETAILS

<u>Sr No.</u>	<u>Roll No.</u>	<u>Name of Student</u>	<u>Enrollment No.</u>	<u>Exam seat No.</u>
1	2	Sarah Shaikh	2207110109	208454
2	10	Siddhi Darekar	23211830367	208467
3	21	Pradnya Kaurav	23211830381	208481
4	66	Subiya Sayyed	23211830431	210904

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1.0 Aim of Micro Project (In about 4 to 5 sentences)

To test and validate all functionalities of an e-commerce website, including user registration, login, product search, shopping cart, checkout, and payment processes. To identify and report bugs or errors to ensure the website is reliable and error-free before deployment.

2.0 Action Plan

Sr. No	Details of activity	Planned start date	Planned Finish date	Name of Responsible Team members
1	Dataset Collection and Study			Sarah Shaikh Siddhi Darekar Pradnya Kaurav Subiya Sayyed
2	Model Development			
3	Report Preparation and Submission			

3.0 Resources Required

Sr. No.	Name of resource	Specification	Quantity	Remarks
1	Hardware	Computer/Laptop with minimum Intel i3 processor, 4GB RAM, and 256GB storage running Windows or Linux.	1	
2	Software	Excel	-	

Abstract

E-commerce websites are widely used platforms that allow users to buy and sell products online. With the increasing use of online shopping, it is important to ensure that these websites function correctly, securely, and efficiently. Software testing plays a crucial role in identifying defects and ensuring that the system meets user requirements.

This project focuses on testing the basic functionalities of an e-commerce website such as user registration, login, product search, add to cart, checkout, and payment process. Various testing techniques like functional testing, usability testing, and performance testing are used to verify the proper working of the system.

The main objective of this project is to detect errors, improve system reliability, and ensure a smooth user experience. Proper testing of e-commerce applications helps in maintaining data security, improving website performance, and increasing customer satisfaction.

- Ensures proper functioning of the e-commerce website
- Tests features like login, product search, cart, and payment
- Helps identify bugs and errors
- Improves website performance and security
- Provides better user experience and reliable online shopping

1.Introduction

E-commerce websites have become an important part of online business, allowing customers to buy and sell products through the internet. With the rapid growth of online shopping, it is necessary to ensure that these websites work properly and provide a smooth user experience.

E-commerce testing is a type of software testing that checks the functionality, performance, security, and usability of an online shopping website. It verifies important features such as user registration, login, product search, shopping cart, checkout process, and online payment.

The main purpose of e-commerce testing is to identify errors and ensure that all features of the website work correctly. Proper testing helps improve system reliability, protect user data, and provide safe and efficient online transactions.

In addition, e-commerce testing also checks whether the website works properly on different browsers and devices. It ensures that customers can easily navigate the website, find products, and complete their purchases without any issues. This helps increase customer satisfaction and trust in the online shopping system

2.Objectives (E-commerce Testing)

- To understand the concept of e-commerce website testing.
- To verify the functionality of features like registration, login, product search, cart, and payment.
- To identify bugs and errors in the system.
- To ensure the website works properly and efficiently.
- To improve security and protect user data.
- To check the performance of the website.
- To ensure the website works on different browsers and devices.
- To provide a smooth and better user experience.
- To make online transactions safe and reliable.

3. Modules of E-Commerce Website Testing

E-commerce testing usually focuses on testing different modules of an online shopping website. Each module is tested to ensure that it works correctly and provides a smooth user experience.

1. User Registration Module

- Allows new users to create an account.
- Checks for valid input fields, email verification, and password security.

2. Login Module

- Allows registered users to log in.
- Tests valid and invalid credentials, password recovery, and session management.

3. Product Search Module

- Allows users to search for products using keywords or categories.
- Tests search accuracy, filters, and sorting options.

4. Product Details Module

- Displays detailed information about a product.
- Tests product images, description, price, reviews, and availability.

5. Add to Cart Module

- Lets users add products to their shopping cart.
- Tests quantity update, removal of items, and cart summary.

6. Checkout Module

- Allows users to place orders for selected products.
- Tests billing/shipping details, order summary, and address validation.

7. Payment Module

- Processes online payment through various payment gateways
- Tests payment success/failure, security, and transaction confirmation.

8. Order Confirmation & Tracking Module

- Confirms the order and provides tracking information.

9. Customer Support Module (Optional)

- Provides help through chat, email, or call support.
- Tests response time and accuracy of support.

10. Performance & Security Module

- Ensures website loads quickly and protects user data.
- Tests for speed, stress, and vulnerability to attacks.

4.Types of Testing in E-Commerce Websites



1. Functional Testing

Functional testing is performed to ensure that all features and functionalities of an e-commerce website work correctly. It checks whether the website allows users to complete important tasks without errors, such as registering an account, logging in, searching for products, adding items to the cart, proceeding to checkout, and placing orders. This type of testing is essential because any failure in core functions can prevent users from making purchases and affect the business negatively.

Examples: User login/logout, Add to Cart, Checkout process.

2. User Interface (UI) Testing

User Interface testing focuses on the visual elements and layout of the website. It ensures that buttons, images, text, menus, and other design components are displayed correctly and work as intended. A clean and responsive interface enhances user experience, helps users navigate easily, and reduces the chance of errors during shopping.

Examples: Clickable buttons, properly aligned text, correct display of product images.

3. Security Testing

Security testing protects the website and its users from unauthorized access, hacking attempts, and data breaches. It ensures that sensitive information, like login credentials, personal details, and payment information, is encrypted and secure. This testing also checks the safety of payment gateways and verifies that the website is free from vulnerabilities like SQL injection or phishing attacks.

Examples: Secure login system, encrypted payment transactions.

4. Performance Testing

Performance testing evaluates the speed, stability, and responsiveness of the website under normal and heavy traffic conditions. It checks how the website performs when multiple users are shopping simultaneously, especially during sales or special promotions. This ensures that the website does not slow down, crash, or fail to respond, which could otherwise lead to a poor shopping experience and loss of sales.

Examples: Page load time, server response during Black Friday sales.

5. Database Testing

Database testing ensures that all data stored in the e-commerce website's database is accurate, consistent, and secure. It verifies that product information, user accounts, order details, and transaction records are correctly saved and retrieved when needed. This type of testing helps prevent data loss, incorrect orders, or payment issues that can negatively impact customers and business operations.

Examples: Checking product details, verifying order history, validating transaction records.

5 E-commerce Website Testing Diagram



6. Test cases E-Commerce Website Testing

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
TC01	User Registration	Enter name, email, password and click register	Account should be created successfully	Account created successfully	Pass
TC02	User Login	Enter email and password and click login	User should login successfully	User logged in successfully	Pass
TC03	Product Search	Enter product name in search bar and click search	Related products should be displayed	Relevant products displayed	Pass
TC04	Add Product to Cart	Select product and click add to cart	Product should be added to cart	Product added to cart successfully	Pass
TC05	Remove Product from Cart	Go to cart and click remove button	Product should be removed from cart	Product removed successfully	Pass
TC06	Checkout	Go to cart and click checkout	User should be redirected to payment page	User redirected to payment page	Pass
TC07	Payment	Select payment method, enter details and confirm	Payment should be completed successfully	Payment processed successfully	Pass
TC08	Order Confirmation	Complete payment and wait for confirmation	Order confirmation message should appear	Order confirmation displayed	Pass
TC09	Apply Coupon	Enter valid coupon code and click apply	Discount should be applied to total amount	Discount applied successfully	Pass
TC10	Logout	Click logout button	User should be logged out successfully	Logged out successfully	Pass

7. Testing Process for E-Commerce Website

1. Requirement Analysis

- Explanation: Study the website's features like login, product search, shopping cart, checkout, and payment to identify what needs to be tested.
- Examples: Verify login validation, check product categories, and confirm checkout flow works correctly.

2. Test Planning

- Explanation: Prepare a detailed test plan covering scope, testing types (functional, UI, security, performance, database), resources, and schedule.
- Examples: Plan functional and security testing, schedule performance tests for high traffic periods.

3. Test Case Design

- Explanation: Create detailed test cases with scenarios, step-by-step actions, expected results, and status fields for Pass/Fail.
- Examples: Test adding/removing items from the cart, applying coupon codes, and checking order confirmation messages.

4. Test Environment Setup

- Explanation: Set up browsers, devices, network conditions, and tools to replicate the production environment accurately.
- Examples: Test on Chrome, Firefox, mobile devices, and simulate slow network or high traffic conditions.

5. Test Execution

- Explanation: Execute all test cases, check each functionality, record actual results, and mark Pass/Fail.
- Examples: Run product search, verify checkout process, complete payment, and check order confirmation.

6. Bug Reporting

- Explanation: Document defects with module name, description, severity, steps to reproduce, and expected vs actual results.
- Examples: Login fails with valid credentials, payment gateway error, broken "Add to Cart" button.

7. Retesting & Regression Testing

- Explanation: Retest fixed defects to confirm they are resolved and perform regression testing to ensure existing features remain unaffected.
- Examples: Verify fixed checkout issue works, ensure product search and cart still function properly.

8. Test Closure

- Explanation: Summarize testing results including total test cases executed, passed, failed, and unresolved bugs.
- Examples: Prepare a test summary report, list critical open bugs, highlight major issues fixed during testing.

Testing Process for E-Commerce Website Diagram



8. Bug Report – E-Commerce Website Testing

Bug ID	Feature	Description	Severity	Expected Result	Actual Result	Status
B001	Registration	Cannot register with valid email	High	Account created successfully	Error message displayed	Open
B002	Login	Login fails with correct credentials	High	User should log in successfully	"Invalid credentials" error	Open
B003	Add to Cart	Product not added sometimes	Medium	Product added to cart	Nothing happens	Open
B004	Payment	Payment page freezes	Critical	Payment processed successfully	Page freezes, not completed	Open
B005	Search	Irrelevant products displayed	Medium	Only related products displayed	Products like "Shoes" shown	Open

9. Payment Gateway in E-Commerce Websites

A payment gateway is a service that processes online payments for e-commerce websites securely. It acts as a bridge between the customer's bank or card provider and the merchant's website, ensuring that financial transactions are authorized, verified, and completed safely.

Key Functions

1. Transaction Processing: Handles credit/debit card, net banking, UPI, wallets, and other payment methods.
2. Authorization & Verification: Confirms that the customer has sufficient funds and validates card or bank details.
3. Encryption & Security: Protects sensitive payment data using encryption (like SSL/TLS) to prevent fraud.
4. Payment Confirmation: Sends success or failure messages to the user and updates the order status in the system.
5. Refund & Reversal Handling: Supports refunds, cancellations, and partial reversals if needed.

Common Issues in Payment Gateway

- Transaction failures even with valid card details.
- Payment page freezing during processing.
- Incorrect amounts charged.
- Discounts or coupons not applied correctly.
- Confirmation messages or emails not received.
-

Importance

- Ensures safe and reliable transactions, which is crucial for building customer trust.
- Directly impacts sales and user satisfaction, as payment failures can lead to cart abandonment.
- Must comply with security standards like PCI DSS to protect user financial data.

Diagram





E-commerce Website Testing

1.0 Brief Description:

The project focuses on testing an E-Commerce Website to ensure it is functional, secure, and user-friendly. The main goal is to verify that all key features, such as user registration, login, product search, shopping cart, checkout, payment gateway, and order confirmation, work correctly.

2.0 Aim of Micro Project

To test and validate all functionalities of an e-commerce website, including user registration, login, product search, shopping cart, checkout, and payment processes, To identify and report bugs or errors to ensure the website is reliable and error-free before deployment.

3.0 Course Outcome Integrated

- a. Select suitable Software Process model for Software development.
- b. Prepare software requirement specification.
- c. Use Software modeling to create data designs.
- d. Estimate size and cost of create data designs.

4.0 Actual Procedure Followed.

1. Selection of topic.
2. Information collected from books
3. Information collected from web resources.
4. Designing algorithm for project.
5. Solving error and testing of project.
6. Preparation of report

Sr. No.	Name of resource	Specification	Quantity	Remarks
1	Hardware	Computer/Laptop with minimum Intel i3 processor, 4GB RAM, and 256GB storage running Windows or Linux.	1	
2	Software	Python 3.x with libraries like Pandas, NumPy, Matplotlib, and Scikit-learn for Software Testing implementation.	-	

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Conclusion

Testing an e-commerce website is a critical process to ensure that all functionalities work correctly and users have a smooth, secure shopping experience. Through requirement analysis, test planning, test case design, test execution, bug reporting, and retesting, the website can be thoroughly evaluated for both functionality and reliability.

By performing functional, UI, performance, security, and database testing, potential issues like login failures, cart errors, payment gateway problems, and UI inconsistencies can be identified and resolved before deployment. Proper testing not only reduces the risk of errors but also enhances customer trust and satisfaction, which is vital for online business success.

Teacher Evaluation Sheet

Name of Student: Sarah Shaikh

Enrollment No: 2207110109

Name of programme: Computer Engineering.

Semester: VI

Course Title: Software Testing

Code: 316314

Title of Micro Project: E-commerce

Website Testing Course Outcomes Achieved:

1. Select suitable Software Process model for Software development.
2. Prepare software requirement specification
- . 3. Estimate size and cost of create data designs

Evaluation as per suggested Rubric for Assessment of Micro Project

Sr. No	Characteristic to be assessed	Poor (Marks1- 3)	Average (Marks 4-5)	Good (Marks6-8)	Excellent (Marks 9-10)
1	Relevance to the course				
2	Literature Survey / Information collection				
3	Project Proposal				
4	Completion of the Target as per Project Proposal				
5	Analysis of data and representation				
6	Quality of Prototype/ Model				
7	Report preparation				
8	Presentation				
9	Defense				

Micro Project Evaluation Sheet

Process Assessment		Product Assessment		Total Marks 10
Part A - Project Proposal (2 Marks)	Project Methodology (2 Marks)	Par B - Project Report/ working Model (2 Marks)	Individual Presentation/ Viva (4 Marks)	

Note: Every course teacher is expected to assign marks for group evaluation in first 3 columns and individual evaluation 4th column

Comment/ suggestion about team work/leadership/ interpersonal communication (If any)

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Any other comment:

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Name and Designation of the Faculty Member:- Ms.S.P.Sarode

Signature:

Teacher Evaluation Sheet

Name of Student: Siddhi Darekar

Enrollment No: 23211830367

Name of programme: Computer Engineering_

Semester: VI

Course Title: Software Testing

Code: 316314

Title of Micro Project: E-commerce

Website Testing Course Outcomes Achieved:

1. Select suitable Software Process model for Software development.
2. Prepare software requirement specification
- . 3. Estimate size and cost of create data designs

Evaluation as per suggested Rubric for Assessment of Micro Project

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Micro Project Evaluation Sheet

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Comment/ suggestion about team work/leadership/ interpersonal communication (If any)

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Any other comment:

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Name and Designation of the Faculty Member:- Ms.S.P.Sarode

Signature:

Teacher Evaluation Sheet

Name of Student: Pradnya Kaurav

Enrollment No: 23211830381

Name of programme: Computer Engineering.

Semester: VI

Course Title: Car Price Prediction

Code: 316314

Title of Micro Project: E-commerce

Website Testing Course Outcomes Achieved:

1. Select suitable Software Process model for Software development.
2. Prepare software requirement specification
- . 3. Estimate size and cost of create data designs

Evaluation as per suggested Rubric for Assessment of Micro Project

Sr. No	Characteristic to be assessed	Poor (Marks1-3)	Average (Marks 4-5)	Good (Marks6-8)	Excellent (Marks 9-10)
1	Relevance to the course				
2	Literature Survey / Information collection				
3	Project Proposal				
4	Completion of the Target as per Project Proposal				
5	Analysis of data and representation				
6	Quality of Prototype/ Model				
7	Report preparation				
8	Presentation				
9	Defense				

Micro Project Evaluation Sheet

Process Assessment		Product Assessment		Total Marks 10
Part A - Project Proposal (2 Marks)	Project Methodology (2 Marks)	Par B - Project Report/ working Model (2 Marks)	Individual Presentation/ Viva (4 Marks)	

Note: Every course teacher is expected to assign marks for group evaluation in first 3 columns and individual evaluation 4th column

Comment/ suggestion about team work/leadership/ interpersonal communication (If any)

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Any other comment:

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Name and Designation of the Faculty Member:- Ms.S.P.Sarode

Signature:

Teacher Evaluation Sheet

Name of Student: Subiya Sayyed

Enrollment No:23211830431

Name of programme: Computer Engineering.

Semester: VI

Course Title: Car Price Prediction

Code: 316314

Title of Micro Project: E-commerce

Website Testing Course Outcomes Achieved:

1. Select suitable Software Process model for Software development.
2. Prepare software requirement specification
- . 3. Estimate size and cost of create data designs

Evaluation as per suggested Rubric for Assessment of Micro Project

Sr. No	Characteristic to be assessed	Poor (Marks1-3)	Average (Marks 4-5)	Good (Marks6-8)	Excellent (Marks 9-10)
1	Relevance to the course				
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Any other comment:

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Name and Designation of the Faculty Member:- Ms.S.P.Sarode

Signature: